

In a class of 140 students numbered 1 to 140, all even numbered students opted Mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is:

- A. 102
- B. 42
- C. 1
- D. 38

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Given that total number of students is 140 numbered 1 to 140

Assume A is the set of students opted for Mathematics also given that even number of students opted for Mathematics so,

$$\begin{aligned} A &= \{2k : k \in Z \text{ and } 2 \leq 2k \leq 140\} \\ &= \{2k : k \in Z \text{ and } 1 \leq k \leq 70\} \end{aligned}$$

Thus, number of elements in A is $n(A) = 70$

Assume B is the set of students opted for Physics also given that the student's having number is divisible by 3 opted Physics so,

$$\begin{aligned} B &= \{3k : k \in Z \text{ and } 3 \leq 3k \leq 138\} \\ &= \{3k : k \in Z \text{ and } 1 \leq k \leq 46\} \end{aligned}$$

Thus, number of elements in B is $n(B) = 46$

Assume C is the set of students opted for Physics also given that the student's having number is divisible by 5 opted Chemistry so,

$$\begin{aligned} C &= \{5k : k \in Z \text{ and } 5 \leq 5k \leq 140\} \\ &= \{5k : k \in Z \text{ and } 1 \leq k \leq 28\} \end{aligned}$$

Thus, number of elements in C is $n(C) = 28$

Now, the students opted for both Mathematics and Physics is represented by the set $A \cap B$ and such students having those numbers divisible by 2 and 3 that is divisible by the 6 (LCM of 2 and 3) so,

$$\begin{aligned} A \cap B &= \{6k : k \in Z \text{ and } 6 \leq 6k \leq 138\} \\ &= \{6k : k \in Z \text{ and } 1 \leq k \leq 23\} \end{aligned}$$

The students opted for both Physics and Chemistry is represented by the set $B \cap C$ and such students having those numbers divisible by 3 and 5 that is divisible by the 15 (LCM of 3 and 5) so,

$$\begin{aligned} B \cap C &= \{15k : k \in \mathbb{Z} \text{ and } 15 \leq 15k \leq 135\} \\ &= \{15k : k \in \mathbb{Z} \text{ and } 1 \leq k \leq 9\} \end{aligned}$$

The students opted for both Mathematics and Chemistry is represented by the set $A \cap C$ and such students having those numbers divisible by 2 and 5 that is divisible by the 10 (LCM of 2 and 5) so,

$$\begin{aligned} A \cap C &= \{10k : k \in \mathbb{Z} \text{ and } 10 \leq 10k \leq 140\} \\ &= \{10k : k \in \mathbb{Z} \text{ and } 1 \leq k \leq 14\} \end{aligned}$$

Therefore, $n(A \cap B) = 23$; $n(B \cap C) = 9$; $n(A \cap C) = 14$

The students opted all three subjects is represented by the set $A \cap B \cap C$ and such students having those numbers divisible by 2, 3 and 5 that is divisible by the 30 (LCM of 2, 3 and 5) so,

$$\begin{aligned} A \cap B \cap C &= \{30k : k \in \mathbb{Z} \text{ and } 30 \leq 30k \leq 120\} \\ &= \{30k : k \in \mathbb{Z} \text{ and } 1 \leq k \leq 4\} \end{aligned}$$

Therefore, $n(A \cap B \cap C) = 4$

Now, the students opted for at least one of the three subjects is represented by $A \cup B \cup C$ so,

$$\begin{aligned} n(A \cup B \cup C) &= n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) \\ &\quad - n(A \cap C) + n(A \cap B \cap C) \\ &= 70 + 46 + 28 - 23 - 9 - 14 + 4 \\ &= 102 \end{aligned}$$

Therefore, the number of students does not obtain any subjects is,

$$= 140 - n(A \cup B \cup C) = 140 - 102 = 38$$

Hence, the **OPTION D** is the correct.